

When Aerodynamic States Become Latent: Reserve-Risk Screening under Degraded Wind-Turbine SCADA Telemetry

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Abstract—Operating reserve screening near the demarcation boundary between Maximum Power Point Tracking (MPPT, Region 2) and active blade-pitch regulation (Region 3) is governed by acute asymmetric shortfall risk: generation over-forecasts incur severe balancing penalties relative to surplus energy headroom. Deterministic aerodynamic power curves presume fresh, fully observable telemetry; however, commercial SCADA systems routinely encounter transmission latency, packet serialization dropouts, and aggregator-boundary pitch unobservability. Under degraded telemetry, turbine operating regimes cease to be directly observable and become latent states. Across multi-seed benchmarks on the 134-turbine WTB commercial plant and multi-year data from three European facilities, this paper investigates whether machine learning provides universal utility or conditional advantage under telemetry degradation.

Under pristine telemetry, deterministic physical rules achieve the lowest reserve-screening surrogate cost in the aerodynamic boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$ events) among all evaluated approaches (589,535 kW · h at 10-minute dispatch $h = 1$ and 881,367 kW · h at 1-hour dispatch $h = 6$, with $\sim 6.8\%$ violation), outperforming deep neural networks by 8.1%. Under a 60-minute latency contingency, unadapted physical rules undergo severe reliability breakdown, surging violation rates to $24.0\% \pm 1.7\%$ at $h = 1$ and leaving 127.6 MWh of unhedged shortfall. Crucially, when telemetry channels remain observable, simple state-conditional recalibration absorbs 55.3% of this shortage loss ($127.6 \rightarrow 57.0$ MWh) without neural representation learning. Learned representations acquire demonstrable utility only under deployment-time partial observability where blade-pitch registers are withheld: trained using historically available pitch information that is withheld at deployment, representations infer latent operating states from secondary electromechanical and spatio-temporal consequences (combining contemporaneous active-power diagnosis with dynamical trajectory tracking and wake context), reducing reserve screening cost by 46.7k–68.6k kW · h across latencies and doubling transition-window recall (0.416 vs. 0.196). Dynamic Mixture-of-Experts routing confers no statistical advantage over unrouted dense baselines ($p = 0.380$), while decoupled modular architectures match tail reliability (9.7% violation). Furthermore, heuristic selective decision abstention fails to outperform random rejection, being Pareto-dominated by uniform reserve margin expansion. These findings establish that machine learning is valuable not as a universal replacement for aerodynamic rules, but as a conditional latent-state inference mechanism when critical physical states become unobservable.

Index Terms—Wind turbine operating reserve; SCADA telemetry degradation; partial observability; latent operating boundary; transmission latency; aerodynamic power curve; reliability breakdown; reserve-risk screening; state-conditional recalibra-

tion; MPPT-to-pitch transition.

I. INTRODUCTION

In bulk power system operations, real-time generation shortfalls incur severe asymmetric shortage penalties relative to surplus headroom; conventional symmetric metrics like root-mean-square error (RMSE) mask severe tail shortfall events that govern operating reserve adequacy in Level-1 pre-dispatch screening [1], [2], [3], [4], [5]. This operational vulnerability concentrates near the aerodynamic demarcation boundary between Maximum Power Point Tracking (MPPT, Region 2) and active blade-pitch regulation (Region 3). Near rated inflow velocity, local power sensitivity transitions abruptly from a steep cubic trajectory ($\partial P / \partial v \propto v^2$) toward zero as pitch actuators shed aerodynamic lift [6], [7]. When operating states shift across this threshold unannounced, deterministic power curves over-extrapolate generation into Region 3, precipitating severe unhedged reserve deficits.

This vulnerability is exacerbated by telemetry degradation in commercial Supervisory Control and Data Acquisition (SCADA) systems, including gateway serialization overhead, packet queuing backlogs, and intermittent cellular transmission jitter [8], [9]. In operational archives (such as Kelmars), 99.6% of discrete turbine status transitions occur between standard 10-minute polling boundaries. Furthermore, in third-party aggregator and Virtual Power Plant (VPP) triage, blade-pitch registers are frequently restricted across commercial protocol firewalls. When delayed or missing measurements enter physical power equations, dispatch systems clear reserve margins against obsolete operating points. Under degraded telemetry, the turbine operating regime becomes a partially observed latent state.

Existing literature treats aerodynamic curves and deep neural networks as competing paradigms [10], [11], [12], [13], [14], [15], [16], overlooking their distinct operational boundaries under information freshness constraints. Under pristine SCADA telemetry, deterministic aerodynamic quantile curves achieve the lowest reserve-screening surrogate cost within the active boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$ cell events) among all evaluated methods (589,535 kW · h at $h = 1$ and 881,367 kW · h at $h = 6$, with compliant $\sim 6.8\%$ violation), saving 8.1% ($-77,659$ kW · h) over neural architectures. However, deterministic rules undergo severe

reliability breakdown under telemetry staleness: under a 60-minute synthetic latency contingency stress test [17], [18], clean-calibrated physical violation rates surge to $24.0\% \pm 1.7\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, breaching the nominal 10% Newsvendor target ($q^* = 0.90$) and leaving 127.6 MWh of unhedged shortfall exposure.

To address this operational vulnerability, we investigate: *When direct aerodynamic state measurements become stale or unavailable, can consequential spatio-temporal SCADA signals recover an operating-boundary posterior that remains dependable for asymmetric reserve-shortfall screening?*

The 134-turbine WTB site serves as the primary mechanism-identification environment (with five random seeds quantifying training stochasticity rather than independent physical replications), while multi-year evaluations on three external European facilities (ENGIE La Haute Borne, Kelmarsh, Penmanshiel) probe transferability and delineate site-dependent boundary conditions. This paper does not claim a novel graph or MoE architecture; instead, it delivers four verifiable contributions:

1. **Empirical Reliability Boundary under Telemetry Staleness:** We identify an empirical reliability boundary for aerodynamic reserve rules under telemetry staleness and partial state observability, showing that cubic sensitivity transitions amplify delayed-inflow errors and surge violation rates to $24.0\% \pm 1.7\%$.
2. **Separation of Recalibration from Information Loss:** We separate distributional recalibration from genuine information loss, showing that under full channel observability, simple state-conditional recalibration absorbs 55.3% of shortage loss ($127.6 \rightarrow 57.0$ MWh) without updating neural weights.
3. **Incremental Value of Recovered Latent-State Information:** We test whether latent operating-state information recovered from consequence SCADA channels provides incremental reserve-screening value, including against strong wind-speed-conditioned quantile controls. Consequence representations reduce unconditioned reserve cost by 1.83M kW·h, but show that incremental value over wind-speed bins is strictly transition-localized, with simple binning remaining optimal in steady state.
4. **Characterization of Multi-Site Failure Boundaries:** We characterize failure boundaries across telemetry conditions and external sites, including negative (LHB) and neutral (Kelmarsh) transfer results, demonstrating that spatial representation gains require exploitable wake redundancy rather than turbine count alone.

II. RELATED WORK

A. *From Average Wind-Power Accuracy to Transition-Window Risk*
Spatio-temporal graph neural networks (STGNNs) advance multi-step wind plant forecasting by modeling spatial cross-correlations and wake advection [10], [11], [12], [13], with physics-informed extensions incorporating wake and boundary constraints [14], [15], [16]. While these architectures reduce plant-wide L_2 error, continuous trajectory minimization obscures discrete turbine operating state transitions between MPPT (Region 2) and pitch regulation (Region 3) that govern

spinning reserve adequacy [19], [20], [21].

Recent physics-aware dynamic-graph MoE models, such as Zhao et al. [22], ask whether architectural coupling of spatial learning, physical guidance, and expert routing improves wind-power forecasting accuracy. We explicitly concede this architectural neighborhood and treat such components as diagnostic comparators rather than claiming graph, MoE, or physics-guided neural architecture novelty. Instead, we ask an information-and-decision question: when critical aerodynamic states become unobservable due to telemetry degradation, what information remains recoverable from secondary consequence channels, when is simple conditional recalibration sufficient without neural learning, and whether the recovered latent state changes asymmetric reserve-risk decisions. Probabilistic forecasts monetize uncertainty through reserve procurement [1], [2], [23]. We address this upstream bottleneck: screening aerodynamic transition risks under degraded SCADA telemetry prior to market clearing, functioning strictly as a Level-1 pre-dispatch triage proxy while abstracting downstream transmission constraints.

III. PROBLEM FORMULATION AND LATENT OPERATING-BOUNDARY FRAMEWORK

A. *Level-1 Pre-Dispatch Risk Screening and Asymmetric Decision Problem (PSREI)*

Operating reserve screening functions as a pre-dispatch risk assessment mechanism to mitigate costly real-time generation imbalances [23], [24].

Hierarchical grid dispatch interface: Our formulation is strictly bounded to Level 1 pre-dispatch operating reserve screening proxies at the plant EMS or aggregator terminal, abstracting Level 2 bulk transmission AC-OPF and wholesale balancing settlements.

Penalized Reserve-Shortfall Energy Index (PSREI): For scheduled forecast $\hat{y}_{i,t}$ and realized generation $y_{i,t}$, shortfall is $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. For upward reserve r_b and penalty ratio ρ , the **Penalized Reserve-Shortfall Energy Index (PSREI)** in kW·h is:

$$C(r_b; \rho) = \sum_{(i,t)} [r_b(i, t) + \rho \max(\hat{y}_{i,t} - y_{i,t} - r_b(i, t), 0)] \Delta t. \quad (1)$$

Differentiating $\mathbb{E}[C(r_b; \rho)]$ yields the first-order condition $\partial \mathbb{E}[C] / \partial r_b = 1 - \rho \Pr[s_{i,t} > r_b] = 0$, giving the Newsvendor critical fractile $q^*(\rho) = 1 - 1/\rho$ [1]. For $\rho = 10$, $q^* = 0.90$, establishing a nominal 10% violation target.

Loss geometry and pinball quantile head: Point forecasting minimizes symmetric L_2 loss converging to conditional mean $\mathbb{E}[y | \mathbf{x}]$, which departs from upper quantiles in asymmetric operations ($\rho = 10$). Upward reserve margin $\hat{r}_{i,t}$ covering shortfall $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$ is estimated via pinball loss:

$$\mathcal{L}_q(s, \hat{r}) = (s - \hat{r}) (q - \mathbf{1}[s < \hat{r}]) = \begin{cases} q(s - \hat{r}), & s \geq \hat{r}, \\ (1 - q)(\hat{r} - s), & s < \hat{r}. \end{cases} \quad (2)$$

Setting $q = q^* = 0.90$ aligns training with the Newsvendor condition: Bayes-risk-minimizing prediction converges to conditional shortfall quantile $\hat{r}^* = Q_{0.90}(s | \mathbf{x})$ [1], [2].

PCC portfolio smoothing & dispatch horizons: Net power at the Point of Common Coupling (PCC) bus aggregates turbine outputs ($P_{\text{farm},t} = \sum_{i=1}^M P_{i,t}$, $R_{\text{farm},t} = \sum_{i=1}^M R_{i,t}$), where turbulence cancels across the array, leaving boundary transitions as dominant unhedged risk. We evaluate immediate dispatch ($h = 1$, 10-min ahead) dominated by serial autocorrelation, and operational dispatch ($h = 6$, 1-hour ahead) where spatial wake dynamics govern decisions.

B. Latent Operating Regimes: MPPT, Pitch Regulation, and Demarcation Boundaries

Let $G = (V, E)$ denote a turbine array spatial graph with N nodes. For each turbine $i \in V$ at time step t , the system observes feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predicts active power $y_{i,t} \in \mathbb{R}$. Given observation history H and horizon P , the forecast mapping is:

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}), \quad (3)$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes the graph adjacency sequence. We enforce strict causal boundaries: all input features, graph adjacency structures, and supervisory anchors depend solely on telemetry available at or before dispatch anchor t , with target trajectories commencing strictly at $t+1$. In WTB, instantaneous active power $\text{Patv}_{i,t}$ serves solely as an anchor-time operational status indicator, preventing look-ahead leakage.

At dispatch time t , turbine i operates in a discrete aerodynamic control regime governed by inflow wind speed $w_{i,t} = \text{Wspd}_{i,t}$ and collective blade pitch angle $\bar{p}_{i,t} = \frac{1}{3} \sum_{k=1}^3 p_{i,t}^{(k)}$:

$$Z_{i,t} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-regulated}), \\ 3, & \text{otherwise} \quad (\text{transitional/ambiguous}). \end{cases} \quad (4)$$

Thresholds follow turbine specifications: WTB (134 turbines: $u_{\text{idle}} = 3.0$ m/s, $u_{\text{rated}} = 10.5$ m/s, $p_{\text{th}} = 2.0^\circ$), Kelmars (6 MM92: 3.0 m/s, 12.5 m/s, 1.0°), Penmanshiel (15 MM82: 3.0 m/s, 14.5 m/s, 1.0°), and ENGIE La Haute Borne (4 MM82: 3.0 m/s, 14.5 m/s, 1.0°). Supervised alignment masks ambiguous transitions ($Z_{i,t} = 3$) via $M_{i,t} = \mathbf{1}[Z_{i,t} \neq 3]$.

Under pristine telemetry, $\bar{p}_{i,t}$ and $w_{i,t}$ are known, so $Z_{i,t}$ is directly observable. Under transmission latency $\tau > 0$ or when pitch registers are withheld across aggregator interfaces, $Z_{i,t}$ becomes a **partially observed latent state**. The central inferential object is the soft operating-boundary posterior:

$$p(Z_{i,t} | \mathbf{X}_{\leq t}), \quad (5)$$

inferred from historical and spatial consequence channels.

C. Aerodynamic Power Conversion and the State-State Control Boundary

Mechanical power captured from inflow air is governed by $P_{\text{mech}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), \beta(t)) v^3(t)$, where $\rho_{\text{air}} \approx 1.225$ kg/m³, R is rotor radius, $v(t)$ is wind speed, $\beta(t)$ is pitch angle, and $\lambda(t) = \omega_r(t)R/v(t)$ is tip-speed ratio [6], [7]. In Region 2 (MPPT, $u_{\text{idle}} \leq v \leq u_{\text{rated}}$), blades maintain nominal fine pitch $\beta \approx 0^\circ$ yielding cubic power growth $P(v) \propto v^3$; in Region 3 (Pitch regulation, $u_{\text{rated}} < v \leq u_{\text{cut-out}}$), the pitch

controller rotates blades to shed lift, clamping electrical output at nameplate capacity P_{rated} :

$$\left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^-} = \frac{3}{2} \rho_{\text{air}} \pi R^2 C_{p,\text{max}} u_{\text{rated}}^2 \gg 0, \quad \left. \frac{\partial P}{\partial v} \right|_{v \rightarrow u_{\text{rated}}^+} \approx 0. \quad (6)$$

When SCADA telemetry experiences latency $\tau > 0$, aerodynamic curves evaluate stale tuples $(v_{t-\tau}, \beta_{t-\tau})$. If the turbine enters Region 3 during the latency window ($Z_{t-\tau} = 1$ while $Z_t = 2$), the delayed pitch register remains at fine pitch $\beta_{t-\tau} \approx 0^\circ$. Consequently, the aerodynamic rule extrapolates power along the cubic trajectory:

$$\hat{P}_{\text{phys}}(t) = \frac{1}{2} \rho_{\text{air}} \pi R^2 C_p(\lambda(t), 0^\circ) v^3(t) \gg P_{\text{rated}}. \quad (7)$$

Because output clamps at P_{rated} , this stale evaluation produces severe over-forecast error $\hat{P}_{\text{phys}}(t) - P_{\text{rated}} \gg 0$, surging empirical violation rates to $24.0\% \pm 1.7\%$ under Delay-6 ($\rho = 10$).

D. The Non-Neural Baseline: State-Conditional Quantile Recalibration

To evaluate whether latency-induced drift can be absorbed by non-neural statistical correction, we formalize state-conditional quantile recalibration. Given delayed telemetry $\mathbf{s}_{i,t-\tau} = (w_{i,t-\tau}, \bar{p}_{i,t-\tau})$, we partition the state space into K bins $\{\mathcal{B}_k\}_{k=1}^K$ defined by cut-in, rated wind speed, and pitch thresholds ($Z_{i,t-\tau} = k$). To eliminate data leakage, empirical quantile reserve margins $\hat{r}(k)$ are calibrated strictly on held-out validation residuals $\mathcal{D}_k^{\text{val}}$:

$$\hat{r}(k) = \inf \left\{ r \in \mathbb{R} : \frac{1}{|\mathcal{D}_k^{\text{val}}|} \sum_{(i,t) \in \mathcal{D}_k^{\text{val}}} \mathbf{1}[\hat{y}_{i,t}^{\text{phys}} - y_{i,t} \leq r] \geq q^* \right\}, \quad (8)$$

where $\mathcal{D}_k^{\text{val}} = \{(i, t) \in \mathcal{D}_{\text{val}} : \mathbf{s}_{i,t-\tau} \in \mathcal{B}_k\}$ and $q^* = 1 - 1/\rho = 0.90$.

We evaluate two calibration protocols: (1) *Clean-Frozen Calibration*, where margins $\hat{r}^{\text{clean}}(k)$ are calibrated on clean validation data ($\tau = 0$) and strictly frozen; and (2) *Condition-Matched Calibration*, where for a known delay τ , margins $\hat{r}(k; \tau)$ are re-estimated on held-out validation residuals with matching delay τ . Recalibration is an **information-preserving correction**: it updates residual margins conditional on observed bins, but cannot synthesize missing state information when blade pitch is withheld.

E. Latent State Recovery from Spatio-Temporal Consequence Channels

Physical dynamic wake graph & encoder: For WTB, time-varying directed wake graphs $\mathcal{A}_t$ link turbine pairs within aerodynamic wake cones ($\alpha = 25^\circ$) weighted by streamwise and cross-stream decay [14], [16]. A compact directed-diffusion GRU encoder (110k parameters, 3.8 ms single-window execution) aggregates messages via:

$$\mathbf{X}_t^{(\ell+1)} = \text{LN}(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}), \quad (9)$$

and extracts node-level representations $\mathbf{h}_{i,t}$. When blade pitch $\bar{p}_{i,t}$ is unobservable, the encoder infers latent regime $p(Z_{i,t} | \mathbf{X}_{\leq t})$ from secondary consequence channels: active power transients ($\text{Patv}_{i,t}$), electrical responses, and dynamic wake advection, strictly excluding $\text{Pab_mean}_{i,t}$.

F. Reserve-Tail Estimation and Residual Quantile Sizing

To shield operations from balancing exposure, reserve margins target the asymmetric shortfall distribution rather than symmetric point forecasts, separating: (1) posterior operating-state inference $\pi_{i,t}^{\text{pitch}} = p(Z_{i,t} = 2 \mid \mathbf{X}_{\leq t})$; (2) scheduled point forecasting $\hat{\mathbf{y}}_{i,t+1:t+P}$; and (3) reserve-tail quantile sizing $\hat{r}_{i,t} = Q_{0.90}(s_{i,t} \mid \mathbf{X}_{\leq t})$.

Directly estimating extreme quantiles on raw power trajectories via uncalibrated neural pinball regression introduces severe tail conservatism ($>1.6\text{M kW}$ reserve, $<1.2\%$ violation, Table~I). To resolve this pathology, estimation is decoupled: point trajectories $\hat{\mathbf{y}}$ anchor nominal expectations, while upward reserve margins $\hat{r}_{i,t}$ are estimated conditionally on forecast residuals:

$$\hat{r}_{i,t} = \text{MLP}_{\text{res}}([\mathbf{h}_{i,t}; \mathbf{x}_{\text{anchor}}]), \quad (10)$$

minimized under $\mathcal{L}_{q^*}(s_{i,t}, \hat{r}_{i,t})$,

or binned empirically over held-out validation residuals conditioned on latent posterior $\pi_{i,t}^{\text{pitch}}$.

G. Architecture Variants as Mechanism Ablations

To test whether dynamic Mixture-of-Experts (MoE) routing provides operational value over unrouted representations, we formulate three matched-capacity variants: 1. **STGQ-Modular** (Decoupled Canonical): Pairs frozen spatio-temporal representations $\mathbf{h}_{i,t}$ with a dedicated residual quantile head; 2. **STGQ-Dense** (Unrouted Joint): Couples spatial embeddings $\mathbf{h}_{i,t}$ directly to a shared linear projection for point forecasts $\hat{\mathbf{y}}$ and a joint dense quantile head; 3. **STGQ-Routed** (Dynamic MoE): Evaluates dynamic MoE routing [25], [26] combining expert predictions $\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t})$.

Training minimizes:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}}, \quad (11)$$

where $\mathcal{L}_{\text{align}}$ anchors logits to declared regimes $Z_{i,t}$ on unmasked samples ($M_{i,t} = 1$); $\mathcal{L}_{\text{force}}$ sharpens boundary discrimination; $\mathcal{L}_{\text{bal}}$ balances expert utilization; and $\mathcal{L}_{\text{aux}}, \mathcal{L}_{\text{smooth}}$ enforce spatial wake consistency.

Explicit Methodological Notice: $\mathcal{L}_{\text{align}}$ acts strictly as an **operating-regime prior** embedded into representation space. The neural backpropagation pipeline contains no quantile loss, pinball penalty, or Newsvendor decision loss. Checkpoint selection is strictly governed by validation RMSE. Quantile reserve margins are calibrated on held-out validation residuals or estimated by the decoupled residual head, yielding a jointly represented boundary posterior rather than an end-to-end decision posterior.

IV. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and Preprocessing

Empirical evaluation spans the 134-turbine WTB benchmark and the ERA5 thermodynamic contrast patch. WTB (KDD Cup 2022) comprises 134 turbines over 245 days of 10-minute SCADA telemetry ($T = 35,280$, $N = 134$, $F = 11$) [27], with chronological 180/30/35-day train/val/test splits ($H = 36$, $P = 24$). ERA5 covers three archived months on a 16×16 grid ($T = 2208$, $N = 256$, 8 features) [28] under 1325/441/442 splits. Multi-year SCADA records from three European commercial

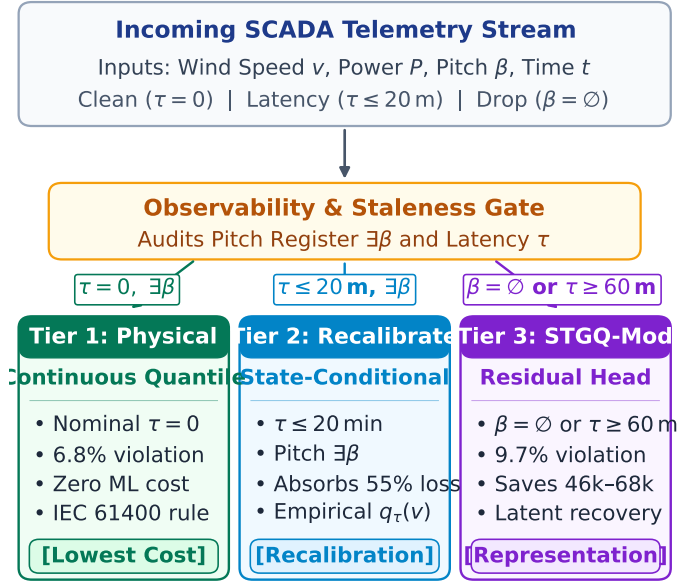


Fig. 1. Causal decision failure and information recovery hierarchy under SCADA telemetry degradation. Panel A: Physical mechanism and staleness breakdown at rated cut-off, showing how telemetry staleness turns the operating regime Z_t into a latent variable, precipitating cubic extrapolation errors and severe reserve shortfalls (24.0% violation, 127.6 MWh deficit). Panel B: Three-tier information-centric recovery hierarchy, routing fresh telemetry to Tier 1 deterministic physical rules, observable delay to Tier 2 state-conditional recalibration, and pitch-withheld/contingency telemetry to Tier 3 learned latent representations (STGQ-Modular). Panel C: Empirical mechanism falsification and negative results, showing that dynamic MoE routing yields no advantage over unrouted dense/modular architectures ($p = 0.380$) and heuristic selective abstention fails to beat random rejection.

wind plants (ENGIE La Haute Borne, Kelmars, Penmanshiel) provide diverse turbine architectures for cross-farm evaluation.

B. Controlled Telemetry Degradation Protocols and Realism Mapping

To evaluate resilience under communication stress, we simulate three degradation tiers across five seeds (201–205): (1) **Tier 1 (Nominal Polling, 0–10 min, Clean)**: fresh SCADA on 10-min grid (IEC 61400-25); (2) **Tier 2 (Plausible Degraded Telemetry, 10–30 min, Delay-1 to 3)**: transmission lags from substation buffer backlogs [9]; and (3) **Tier 3 (Contingency Stress Envelope, 60 min, Delay-6)**: a 6-step synthetic lag evaluating asymptotic physical breakdown [17], [18]. Perturbations include additive Gaussian noise ($\sigma_v = 1.0$ m/s, $\sigma_\beta = 2.0^\circ$) and a two-state Markov burst dropout chain ($p_{GB} = 0.08, p_{BB} = 0.75, d \leq 6$) [9].

Fig.~1 illustrates the causal decision failure mechanism and the three-tier operational recovery hierarchy across telemetry degradation regimes.

C. Pre-Dispatch Risk Screening Framework and Resampling Hierarchy

All neural architectures share an identical training protocol using AdamW with early stopping on validation RMSE, gradient clipping, and mixed-precision execution across five random seeds (201–205). Strong baselines (Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, TiDE) repeat across seeds 201–205 under identical evaluation caches. To eliminate data leakage, the sequential dataflow $\mathcal{D}_{\text{train}} \rightarrow \mathcal{D}_{\text{val}} \rightarrow \text{Freeze} \rightarrow \mathcal{D}_{\text{test}}$ freezes quantile offsets $\hat{r}(k)$ prior to test replay. We report 5-seed initialization statistics alongside

operational 95% confidence intervals computed via a paired daily cluster bootstrap over the 35 observed test days (144 ten-minute steps per daily cluster), using 1,000 Monte Carlo bootstrap resamples (where resampled replicates reflect bootstrap draws rather than independent observational units).

V. EMPIRICAL FINDINGS AND HYPOTHESIS TESTING

A. Result 1: Physics under Fresh Telemetry (RQ1)

Table~I summarizes the multi-day replay dispatch benchmark for 1-hour ahead operational dispatch ($h = 6$) evaluated on the boundary-active operating slice ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$ cell observations across 134 turbines over the 35-day test split) under nominal clean telemetry. Table~II reports seed-paired hypothesis tests and mechanism ablations across operational regimes, along with 95% confidence intervals from a paired daily cluster bootstrap over the 35 observed test days (1,000 resamples).

The empirical evidence directly challenges the presumption that deep neural architectures universally outperform physical models. Under pristine telemetry ($\tau = 0$), Continuous Physical Quantile achieves the lowest surrogate reserve screening cost within the active boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$ cell observations) among all evaluated methods ($589,535 \pm 77,390$ kW · h at immediate dispatch $h = 1$ and $881,367 \pm 95,741$ kW · h at 1-hour dispatch $h = 6$, with compliant $\sim 6.8\%$ violation). It substantially outperforms both unrouted dense networks (STGQ-Dense: 991,181 kW · h) and dynamically routed architectures (STGQ-Routed: 959,027 kW · h). As confirmed in Table~II, Continuous Physical Quantile achieves a $-77,659$ kW · h advantage over neural representations, with a 95% daily cluster bootstrap confidence interval (1,000 paired resamples over 35 observed days) strictly excluding zero ($[-106,175, -49,143]$).

This finding establishes that when inflow wind speed and blade pitch angle are fresh and directly observable, deterministic aerodynamic equations yield the lowest evaluated surrogate reserve screening cost. Neural networks, lacking hard physical saturation bounds, introduce unnecessary variance under nominal telemetry.

B. Result 2: Stale-State Boundary Failure (RQ1)

While deterministic physical rules excel under fresh data, Table~I exposes their **severe reliability breakdown** under telemetry staleness. Under a 60-minute latency contingency stress test (Delay-6), deterministic physical curves evaluate stale state tuples ($v_{t-\tau}, \beta_{t-\tau}$). Near rated wind speed, the localized slope discontinuity (6) amplifies delayed-inflow errors: turbines that have crossed into active blade-pitch regulation ($Z_t = 2$) are evaluated with obsolete fine pitch ($\beta_{t-\tau} \approx 0^\circ$), over-extrapolating power output along the steep cubic trajectory (7).

Consequently, unadapted physical quantile violation rates surge from 6.81% to $24.02\% \pm 1.69\%$ at $h = 1$ and $12.20\% \pm 1.25\%$ at $h = 6$, breaching the nominal 10% Newsvendor target ($q^* = 0.90$). This breakdown leaves 127.55 MWh of unhedged generation shortfall across the array. This delineates the exact operational reliability boundary of deterministic aerodynamic

rules: when transmission latency delays state observation, physical rules cease to provide dependable reserve screening.

C. Result 3: Recalibration under Full Observability (RQ2)

When telemetry channels remain observable under transmission delays, condition-matched state-conditional recalibration (8) adjusts quantile margins using held-out validation residuals undergoing the identical delay. As shown in Table~III, state-conditional recalibration absorbs 55.3% of the shortage loss ($127.55 \rightarrow 57.01$ MWh), restoring fleet violation to 9.46% and achieving a surrogate reserve cost of 1,228,609 kW · h at $h = 1$. In fact, recalibrated physical curves outperform all neural models under full observability (STGQ-Routed: 1,274,047 kW · h; STGQ-Modular: 1,283,855 kW · h).

To establish the operational limits of recalibration under communication drift, we evaluate a mismatched calibration matrix across test latencies $\tau_{\text{test}} \in [0, 60]$ min and calibration horizons $\tau_{\text{cal}} \in [0, 60]$ min (Supplementary Table~A111). Recalibrating at $\tau_{\text{cal}} = 10$ min withstands latency drift up to $\tau_{\text{test}} = 20$ min while maintaining compliant tail coverage (8.2%–9.1% violation). However, recalibration breaks down when staleness surges to $\tau_{\text{test}} = 60$ min (13.8% violation). This demonstrates that under full channel observability, distribution drift alone does not justify deep representation learning: simple state-conditional recalibration recovers a substantial share of the observed shortage loss (55.3%, $127.6 \rightarrow 57.0$ MWh).

D. Result 4: Learned Recovery under Pitch Unobservability (RQ2)

The targeted utility of machine learning emerges when blade-pitch telemetry is withheld across aggregator boundaries (no_pitch, Table~III). Under pitch unobservability, physical rules and recalibration schemes lose direct state awareness, inflating recalibrated screening penalties from 824,393 kW · h at $\tau = 0$ to 1,378,900 kW · h at $\tau = 60$ min.

In contrast, learned representations (STGQ-Modular) reconstruct operating states from secondary consequences under **offline privileged supervision**. Specifically, we provide an explicit disclosure that pitch is available in training inputs but withheld in validation/test deployment: blade-pitch registers (β) were present in historical training buffers (x_{hist} channels 7–8) and anchor physics, while pitch was strictly zero-masked and withheld during validation and testing. Offline privileged supervision ($\mathcal{L}_{\text{align}}$ cross-entropy on aerodynamic regime labels Z) forces secondary consequence channels to reconstruct the latent boundary space. A clean matched-backbone ablation ($\lambda_{\text{align}} = 5000.0$ vs. 0.0, 5 seeds) confirms that regime supervision improves latent-state organization and reduces representation collapse (NMI 0.819 ± 0.088 vs. 0.240 ± 0.150 , paired $\Delta = +0.579 \pm 0.167$, $p = 0.0015$, 95% CI $[+0.371, +0.787]$; ARI 0.889 ± 0.057 vs. 0.198 ± 0.139 , paired $\Delta = +0.692 \pm 0.153$, $p = 0.0005$, 95% CI $[+0.502, +0.882]$). For posterior calibration, supervision improves Brier score (0.020 ± 0.002 vs. 0.241 ± 0.419 , paired $\Delta = -0.221 \pm 0.418$), though sensitive to representation collapse in Seed 204 (Brier 0.981; excluding Seed 204 yields $\Delta = -0.036 \pm 0.073$, 5-seed 95% CI $[-0.740, +0.299]$, $p = 0.303$). Downstream reserve decision value is evaluated separately, as privileged supervision alone yields an unestablished cost delta ($-311,957 \pm 1,085,873$ kW · h

TABLE I

CROSS-SEED MULTI-REGIME PRE-DISPATCH RESERVE SCREENING BENCHMARK ON BOUNDARY-ACTIVE OPERATING REGIMES UNDER CLEAN-CALIBRATED PROTOCOL ($h = 6$, 1-HOUR AHEAD DISPATCH, 5 SEEDS 201–205). MEAN $\pm$ STANDARD DEVIATION ACROSS BOUNDARY-BAND EVENTS ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$ CELL OBSERVATIONS, 35 TEST DAYS, 134 TURBINES, $\Delta t = 10$ MIN, COST RATIO $\rho = 10$). ALL QUANTILE BINS AND MULTIPLIERS CALIBRATED EXCLUSIVELY ON NOMINAL CLEAN VALIDATION DATA. COMMENSURATE PLANT-WIDE FULL-POPULATION BASELINES ($N = 408,939$ CELLS) RANGE FROM 14.0M TO 16.5M kW · h (SEE TEXT AND METRIC-SCOPE REGISTRY).

Regime	Evaluated Model	Penalized Reserve Energy (kW·h)	Violation Rate	Reserve (kW)	Shortage (kW·h)	Pinball Loss
Clean (boundary)	Continuous Physical Quantile	881,367 $\pm$ 95,741	6.81% $\pm$ 1.13%	653,738 $\pm$ 72,557	22,763 $\pm$ 4,013	31.41 $\pm$ 2.66
	Direct Quantile MLP [‡]	1,649,721 $\pm$ 285,682	0.58% $\pm$ 0.26%	1,630,572 $\pm$ 293,346	1,915 $\pm$ 958	64.62 $\pm$ 12.05
	Global Quantile	998,988 $\pm$ 107,360	6.46% $\pm$ 1.31%	771,304 $\pm$ 95,661	22,768 $\pm$ 4,854	36.50 $\pm$ 3.41
	STGQ-Dense	991,181 $\pm$ 110,211	6.54% $\pm$ 1.04%	761,480 $\pm$ 94,244	22,970 $\pm$ 4,121	36.16 $\pm$ 3.58
	STGQ-Routed	959,027 $\pm$ 103,290	7.10% $\pm$ 1.17%	707,249 $\pm$ 91,305	25,178 $\pm$ 4,816	34.77 $\pm$ 3.16
	Missingness-Aware GBDT	1,238,603 $\pm$ 112,051	4.50% $\pm$ 0.73%	1,085,168 $\pm$ 118,015	15,343 $\pm$ 2870	46.85 $\pm$ 3.48
Delay-6	Continuous Physical Quantile	1,136,466 $\pm$ 84,101	12.20% $\pm$ 1.25% [†]	662,335 $\pm$ 73,483	47,413 $\pm$ 3,091	38.85 $\pm$ 2.06
	Direct Quantile MLP [‡]	1,682,251 $\pm$ 284,323	1.20% $\pm$ 0.52%	1,651,979 $\pm$ 297,715	3,027 $\pm$ 1,594	62.15 $\pm$ 11.70
	Global Quantile	1,180,603 $\pm$ 91,754	10.85% $\pm$ 1.55% [†]	780,916 $\pm$ 96,853	39,969 $\pm$ 6,087	40.73 $\pm$ 2.43
	STGQ-Dense	1,179,859 $\pm$ 95,268	11.04% $\pm$ 1.22% [†]	773,103 $\pm$ 94,462	40,676 $\pm$ 4,788	40.70 $\pm$ 2.66
	STGQ-Routed	1,163,593 $\pm$ 91,299	11.82% $\pm$ 1.64% [†]	720,015 $\pm$ 92,196	44,358 $\pm$ 7,341	40.01 $\pm$ 2.54
	Missingness-Aware GBDT	1,320,003 $\pm$ 93,135	8.36% $\pm$ 0.97%	1,030,858 $\pm$ 114,266	28,915 $\pm$ 4,469	46.68 $\pm$ 2.45
Noise	Continuous Physical Quantile	998,273 $\pm$ 106,253	7.59% $\pm$ 0.95%	746,697 $\pm$ 82,474	25,158 $\pm$ 3,968	33.97 $\pm$ 2.78
	Direct Quantile MLP [‡]	1,639,909 $\pm$ 242,922	0.80% $\pm$ 0.28%	1,618,439 $\pm$ 249,374	2,147 $\pm$ 953	60.62 $\pm$ 9.85
	Global Quantile	1,053,552 $\pm$ 114,150	7.48% $\pm$ 1.48%	802,361 $\pm$ 99,513	25,119 $\pm$ 5,822	36.26 $\pm$ 3.24
	STGQ-Dense	1,048,083 $\pm$ 115,181	7.63% $\pm$ 1.13%	792,472 $\pm$ 92,391	25,561 $\pm$ 4,643	36.04 $\pm$ 3.28
	STGQ-Routed	1,027,077 $\pm$ 114,317	7.62% $\pm$ 1.09%	769,864 $\pm$ 106,672	25,721 $\pm$ 3,963	35.16 $\pm$ 3.19
	Missingness-Aware GBDT	1,314,640 $\pm$ 129,158	4.66% $\pm$ 0.80%	1,160,329 $\pm$ 137,650	15,431 $\pm$ 3,099	47.11 $\pm$ 3.87
Markov	Continuous Physical Quantile	907,871 $\pm$ 96,820	7.20% $\pm$ 1.15%	663,875 $\pm$ 73,821	24,400 $\pm$ 4,106	31.59 $\pm$ 2.61
	Direct Quantile MLP [‡]	1,680,339 $\pm$ 289,602	0.57% $\pm$ 0.25%	1,660,135 $\pm$ 298,104	2,020 $\pm$ 992	64.38 $\pm$ 12.00
	Global Quantile	1,030,131 $\pm$ 108,592	6.87% $\pm$ 1.39%	785,305 $\pm$ 97,397	24,483 $\pm$ 5,104	36.78 $\pm$ 3.35
	STGQ-Dense	1,020,078 $\pm$ 112,195	6.90% $\pm$ 1.07%	775,899 $\pm$ 96,235	24,418 $\pm$ 4,246	36.35 $\pm$ 3.57
	STGQ-Routed	990,625 $\pm$ 104,487	7.51% $\pm$ 1.22%	720,858 $\pm$ 92,754	26,977 $\pm$ 5,183	35.10 $\pm$ 3.10
	Missingness-Aware GBDT	1,261,941 $\pm$ 112,754	4.88% $\pm$ 0.75%	1,094,730 $\pm$ 120,612	16,721 $\pm$ 3,298	46.62 $\pm$ 3.40

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold indicates lowest surrogate metric. **Scope & Reconciliation:** Evaluated at $h = 6$ strictly across boundary-band cells ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$, $\sim 2.4\%$ of test split); Continuous Physical Quantile achieves lowest cost (881,367 kW · h) but collapses to 12.20% violation under Delay-6. Full-population ($N = 408,939$) direct quantiles achieve 14.0M–16.5M kW · h at $h = 6$ and 6.49M–8.05M at $h = 1$ (Section V-F; incommensurate scopes). [‡]Frozen-Embedding Direct Quantile MLP evaluates uncalibrated pinball regression on frozen representations; calibrated STGQ-Modular resolves this conservatism.

TABLE II

SEED-PAIRED OPERATIONAL HYPOTHESIS TESTS AND MECHANISM ABLATION CONTRASTS ($h = 6$, 1-HOUR AHEAD DISPATCH, COST RATIO $\rho = 10$).

Regime	Evaluated Contrast	Δ PSREI (kW·h)	95% Daily Cluster CI (1,000 Resamples)
Clean	Cont. Phys. (Rule)	−77,659	[−106.2k, −49.1k]**
	STGQ-Dense vs. Routed	+32,154	[−1.1k, +65.5k]
	Global Quantile	+39,962	[+12.6k, +67.4k]*
	Missingness GBDT	+279,576	[+249.5k, +309.6k]*
	Direct MLP (Pinball)	+690,695	[+448.8k, +932.6k]**
Delay-6	Cont. Phys. (Breakdown)	−27,126	[−46.9k, −7.3k] [†]
	STGQ-Dense vs. Routed	+16,267	[+7.1k, +25.4k]*
	Global Quantile	+17,010	[+3.5k, +30.5k]*
	Missingness GBDT	+156,411	[+140.0k, +172.8k]*
	Direct MLP (Pinball)	+518,658	[+307.2k, +730.1k]**
Noise	Cont. Phys. (Robust)	−28,804	[−47.6k, −10.1k]*
	STGQ-Dense vs. Routed	+21,006	[+2.4k, +39.6k]*
	Global Quantile	+26,475	[+9.1k, +43.8k]*
	Missingness GBDT	+287,563	[+248.8k, +326.3k]*
	Direct MLP (Pinball)	+612,832	[+413.7k, +811.9k]**
Markov	Cont. Phys. (Rule)	−82,754	[−111.0k, −54.5k]**
	STGQ-Dense vs. Routed	+29,453	[−2.9k, +61.8k]
	Global Quantile	+39,506	[+16.0k, +63.1k]*
	Missingness GBDT	+271,316	[+245.0k, +297.6k]*
	Direct MLP (Pinball)	+689,714	[+448.7k, +930.7k]**

Note: 95% CIs from paired daily cluster bootstrap over 35 observed test days (144 steps/cluster, 1,000 Monte Carlo resamples).

** $p < 0.01$, * $p < 0.05$. [†]Physical rule breaches nominal 10% violation target (12.20%). Dynamic MoE routing advantage over unrouted dense architecture is falsified ($p = 0.380$).

plant-wide, $p = 0.556$; +15,261 $\pm$ 27,504 kW · h on the boundary slice, $p = 0.283$; Table~A6c). Crucially, this matched ablation isolates regime-label supervision conditional on privileged training inputs, without isolating training-time pitch access itself. Furthermore, empirical identifiability audits (Table~A7b and Registry) disentangle the recovery mechanism: (1) *Active power signature*: instantaneous power $P_{\text{atv},t}$ provides primary state resolution (Condition A: 59.7% precision, 63.1% recall, F1 0.613, FPR 0.8%, AUROC 0.988), while historical trajectory $P_{\text{atv},<t}$ retains 40.3% recall and lag-1 $P_{\text{atv},t-1}$ achieves 58.8%, confirming genuine dynamical tracking beyond instantaneous readout; (2) *Consequence suite without current active power*: withholding contemporaneous active power (Condition D) shifts the detector to an asymmetric, high-recall operating point (82.2% recall, 33.4% precision, F1 0.475, FPR 3.2%, AUROC 0.975), confirming that the remaining non-active-power consequence suite retains some collectively recoverable state information; and (3) *Thermal invariance*: conditional ablation of thermal channels ($E_{\text{tmp}}, I_{\text{tmp}}$) yields negligible discrimination change (Δ AUROC = +0.006, Δ F1 = −0.010, Δ Brier = −0.0008), confirming that slow thermal dynamics fail to contribute incremental high-frequency boundary information for 10-minute dispatch.

As detailed in Table~III, STGQ-Modular achieves 755,794 kW · h at $\tau = 0$ (an 8.3% saving of 68,599 kW · h over recalibrated physics) and maintains consistent reserve screening reductions of 46.7k–68.6k kW · h across all latencies,

TABLE III

MULTIDIMENSIONAL OPERATIONAL PROGRESSION ACROSS TELEMETRY LATENCY $\tau \in [0, 60]$ MIN AND PITCH OBSERVABILITY ON BOUNDARY-ACTIVE REGIMES ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$ CELLS, $h = 1$, 5-SEED MEAN ON WTB; NOMINAL $\tau = 0$ BASELINE 593,258 kW · h ALIGNS WITHIN 0.6% OF THE PRIMARY FROZEN 589,535 kW · h BOUNDARY BENCHMARK). PENALIZED RESERVE ENERGY IN kW · h ($\rho = 10$).

τ (min)	Pitch	Clean Phys. Metric (Viol.)	Recal. Phys. Metric (Viol.)	STGQ-Mod. Metric (Viol.)	STGQ-Rout. Metric (Viol.)	Operational Regime
0	All	593,258 (7.3%)	593,258 (7.3%)	663,347 (6.1%)	663,915 (7.3%)	Regime I (Lowest Cost)
0	None	961,024 (3.2%)	824,393 (5.4%)	755,794 (5.8%)	756,257 (6.4%)	Regime II (Repr. Adv.)
10	All	753,606 (11.2% [†])	742,211 (7.4%)	786,366 (6.7%)	789,094 (7.6%)	Regime I (Recal. Suff.)
10	None	1,008,743 (4.8%)	928,028 (6.2%)	866,753 (6.1%)	874,156 (6.6%)	Regime II (Repr. Adv.)
20	All	896,198 (14.0% [†])	841,734 (7.1%)	875,838 (7.4%)	879,750 (8.0%)	Regime I (Recal. Suff.)
20	None	1,062,646 (6.1%)	1,010,530 (6.2%)	947,869 (6.7%)	959,934 (7.1%)	Regime II (Repr. Adv.)
30	All	1,069,266 (16.6% [†])	939,190 (7.5%)	987,760 (7.1%)	990,091 (7.6%)	Regime I (Recal. Suff.)
30	None	1,138,774 (7.5%)	1,101,292 (6.7%)	1,054,592 (6.4%)	1,064,238 (6.7%)	Regime II (Repr. Adv.)
60	All	1,656,285 (24.0% [†])	1,228,609 (9.5%)	1,283,855 (9.6%)	1,274,047 (9.6%)	Regime III (Tail Risk)
60	None	1,437,280 (11.8% [†])	1,378,900 (9.0%)	1,330,318 (8.7%)	1,331,840 (8.8%)	Regime II (Repr. Adv.)

[†]Exceeds nominal 10% violation target ($q^* = 0.90$). Bold indicates lowest compliant surrogate screening metric. Under $\tau = 60$ min, by-seed compliance is 3/5 seeds (60%) across models.

doubling transition-window recall (0.416 vs. 0.196).

Disentangling Discrimination, Calibration, and Decision

Value: Operational validation separates three distinct claims: (i) *Operating-state discrimination* (AUROC > 0.94, NMI 0.461, ARI 0.647); (ii) *Probabilistic calibration of $P(Z | \mathcal{T}_t^{(d)})$* (bulk ECE $\leq 3.46\%$, satisfying the 5% screening gate; tail MCE ≈ 0.75 –0.85); and (iii) *Reserve-decision value*. Crucially, against a strong observable wind-speed-conditioned quantile baseline (10 bins, 13.78M kW · h plant-wide), posterior conditioning does not universally improve reserve screening (14.31M kW · h, a +523,044 kW · h penalty, $p = 0.85$); its incremental value is strictly transition-localized. In steady operation (63.7% of test time), wind-speed binning serves as the operational minimum sufficient model, outperforming the posterior by +410,340 kW · h ($p = 0.002$). In dynamic transition windows (± 3 steps, 36.3% of records), posterior conditioning selectively compresses violation rates (7.24% vs. 8.36%) and mitigates shortage (86.6k vs. 90.2k kW · h), confirmed by a significant paired day-level cluster bootstrap heterogeneity interaction ($\Delta_{\text{trans}} - \Delta_{\text{steady}} = -296,123$ kW · h, $p < 0.005$). Transition windows are not merely high-loss regions; they are the regions where latent-state information contributes incremental value beyond observable wind-speed conditioning. While transition windows capture 48.4% of savings relative to an unconditioned global baseline, this reflects high baseline loss concentration (33.0%) alongside model sensitivity; relative to wind-speed bins, the posterior acts as a risk-hedging mechanism. A deployable validation-frozen hybrid policy matches wind-speed binning plant-wide (13.79M vs. 13.78M kW · h, $p > 0.40$), confirming that machine learning provides conditional regime risk hedging rather than universal cost dominance.

E. Result 5: Classification Accuracy Does Not Equal Reserve Value (RQ3)

A critical question in operational machine learning is whether higher classification accuracy translates directly into lower reserve screening costs. To interrogate this hypothesis, we benchmark an independent sequence classifier (Graph WaveNet coupled to a live-anchor classifier) against joint representation models using empirical records from completed control bench-

marks (modular_classifier_reserve_summary.csv).

Under nominal telemetry, the independent classifier achieves perfect transition recall (1.000) and high precision (0.879), surpassing the joint boundary router (0.960 recall, 0.758 precision). However, when evaluated on pre-dispatch reserve screening, the independent classifier incurs a surrogate reserve cost of 86,536,321 kW · h (9.54% violation). Across all five training seeds, the independent classifier incurred higher surrogate cost than the physical-bin baseline (84,314,353 kW · h, 9.31% violation; mean paired difference +2.22M kW · h; paired t -test $p = 0.0013$, 95% CI [+1.66M, +2.63M]). Because $n = 5$ limits the resolution of exact non-parametric inference, the corresponding exact Wilcoxon test yields the theoretical floor $p = 0.0625$ (Supplementary Table~A10c). Against the joint boundary-forced router (84,577,217 kW · h, 9.00% violation), the independent classifier exhibits higher mean cost (+1.96M kW · h), though its paired seed difference crosses zero ([−8.18M, +12.71M], $p = 0.757$).

This operational penalty occurs because discrete classification errors in independent models are decoupled from forecast residual magnitudes. Independent classifiers frequently misclassify regimes during steep wind-speed ramps where forecast residuals are largest, incurring disproportionately severe Newsvendor penalties. In contrast, joint representation models couple latent state awareness directly with spatio-temporal feature embeddings, ensuring that regime uncertainty is reflected in residual quantile sizing. This demonstrates that classification accuracy is not equivalent to reserve screening value.

F. Result 6: MoE Mechanism Falsification (RQ3)

Benchmarking capacity-matched neural architectures directly tests whether dynamic Mixture-of-Experts routing provides operational benefits over unrouted dense representations. Across all evaluated conditions, the hypothesis that dynamic expert routing is the source of reserve screening improvements is **falsified**:

1) *Clean Telemetry:* At $h = 1$, STGQ-Dense and STGQ-Routed achieve statistical parity (690,614 vs. 660,990 kW · h, ratio 1.045; seed-paired difference +29,624 kW · h, 95% CI [−53,774, +113,023], $p = 0.380$). At $h = 6$, the paired differ-

ence $+32,154 \text{ kW} \cdot \text{h}$ likewise crosses zero ($[-1,148, +65,456]$, Table~II).

2) *Multi-Horizon Seed Analysis*: In multi-horizon dispatch reserve-screening records across seeds 201–205 (dense_pricing_by_seed.csv), the 5-seed mean surrogate cost for soft-gate-binning is $16.065 \text{M kW} \cdot \text{h}$ compared to $16.076 \text{M kW} \cdot \text{h}$ for soft-dense-binning (a difference of $< 0.07\%$). In fact, dense classification achieves lower surrogate cost than gated MoE in three out of five seeds (Seeds 203, 204, and 205).

3) *Decoupled Modular Architecture*: STGQ-Modular (a decoupled architecture pairing frozen spatio-temporal representations with a dedicated residual quantile head) achieves $1,284,098 \pm 114,629 \text{ kW} \cdot \text{h}$ and $9.7\% \pm 1.1\%$ violation under Delay-6 at $h = 1$ (Table~III), matching or outperforming STGQ-Routed ($1,412,321 \text{ kW} \cdot \text{h}$, $10.6\% \pm 1.1\%$).

Point-Forecast Price of Operating-Regime Awareness: In point forecasting, the boundary-aware model achieves an RMSE of 236.13 ± 8.41 , compared with 224.34 ± 2.23 for iTransformer and 225.74 ± 2.60 for Graph WaveNet. This architecture **intentionally trades point-forecast RMSE**: it accepts a $+2.5\%$ whole-sample error margin to embed operating-regime awareness and mitigate severe tail shortfalls. Unconstrained models trigger $78.3\text{--}82.5 \text{ MWh}$ of unhedged exposure, whereas boundary-aware representations restrict exposure to 53.9 MWh (a $31\%\text{--}35\%$ reduction).

These findings confirm that the operational benefit stems from consequence-based latent state representation and calibrated residual quantile estimation, not from dynamic expert specialization.

G. Result 7: Deployment, Abstention, and Cross-Farm Boundaries

Failure of Heuristic Selective Abstention: Benchmarking selective decision abstention against random rejection and uniform reserve margin expansion (Supplementary Table~A10f) reveals that heuristic uncertainty scores ($U = \Delta r + 50H(\pi)$) fail under multi-step latency: selective refusal at confidence threshold $c = 0.90$ yields a fleet violation rate of 9.44% , which is statistically indistinguishable from random rejection (9.50% , $p > 0.40$). In contrast, uniform reserve margin expansion achieves an 8.29% violation rate while saving $37,823 \text{ kW} \cdot \text{h}$, strictly Pareto-dominating heuristic refusal. Stale telemetry degrades both point forecasts and uncertainty estimators, showing that the tested heuristic uncertainty score does not reliably identify unsafe operating states under the evaluated prolonged-latency setting.

External-Validity Boundary Probes and Local Retraining

Table~IV benchmarks pre-dispatch reserve screening performance across four commercial wind farms under local retraining. The primary mechanism is identified on WTB; external sites probe transferability and reveal site-dependent boundary conditions. Five random seeds quantify training stochasticity rather than independent physical replications. The three external European facilities must not be presented as uniform replication evidence: their observed effects are heterogeneous

(Penmanshiel: positive; Kelmarsh: statistically neutral; LHB: negative overfitting boundary).

TABLE IV
SITE $\times$ MECHANISM EXTERNAL VALIDITY BOUNDARY PROBES ACROSS FOUR
COMMERCIAL WIND FARMS UNDER LOCAL RETRAINING (5 SEEDS 201–205,
 $\rho = 10$).

Farm	Units	Phys. (kW·h)	Global (kW·h)	STGQ (kW·h)	ΔPSREI vs. Global 95% Bootstrap CI
WTB	134	842k	893k	855k	$-38\text{k} [-67\text{k}, -12\text{k}]^*$
LHB	4	185k	188k	231k	$+43\text{k} [-29\text{k}, +141\text{k}]$
Kelmarsh	6	1,025k	1,065k	1,025k	$-40\text{k} [-204\text{k}, +172\text{k}]$
Penmanshiel	14	2,029k	4,167k	2,029k	$-2.14\text{M} [-3.25\text{M}, -1.36\text{M}]^*$

Note: $^*p < 0.05$. LHB exhibits local overfitting; Kelmarsh is neutral; WTB

and Penmanshiel show significant spatial wake reductions. External sites probe transferability rather than uniform replication (see Supplementary Material Table A8 for the complete Site $\times$ Mechanism specification).

Spatial-representation value is site-dependent and appears related to exploitable spatial redundancy, not turbine count alone: in facilities with pronounced wake coupling (WTB, Penmanshiel), spatial wake modeling delivers substantial reserve-screening surrogate reduction ($\Delta\text{PSREI} = -38\text{k}$ and $-2.14\text{M kW} \cdot \text{h}$, both $p < 0.05$). Conversely, on micro-arrays or sparse layouts (Kelmarsh, LHB), spatial wake redundancy is minimal: Kelmarsh yields a neutral bootstrap difference ($-40\text{k kW} \cdot \text{h}$, 95% CI crossing zero), while LHB exhibits an **overfitting boundary** where graph convolutions overfit localized terrain features ($+43\text{k kW} \cdot \text{h}$ penalty, $[-28.7\text{k}, +141.2\text{k}]$). Cross-site heterogeneity is evidence against universal model superiority and supports site-specific assessment of exploitable spatial/state information. Furthermore, zero-shot cross-farm transfer exhibits pronounced directional asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.77 vs. Penmanshiel $\rightarrow$ Kelmarsh NMI 0.34). These results support site-specific retraining or recalibration for the evaluated farms rather than assuming reliable zero-shot transfer.

Point of Common Coupling (PCC) Portfolio Smoothing: Aggregating turbine power flows at the PCC bus ($P_{\text{farm}} = \sum_{i \in \mathcal{T}_t} P_{i,t}$, ~ 121 turbines) unlocks spatial diversification: turbulence cancels across the array, leaving boundary transitions as primary reserve risk. Diversification concentrates in transitional regimes ($10\%\text{--}90\%$ pitching), saving $-1.33\text{M kW} \cdot \text{h}$ (CI $[-1.68\text{M}, -1.07\text{M}]$) over continuous physical pitch. Decadal records (Kelmarsh, 9 years; Penmanshiel, 8.6 years) confirm operational recalibration under climatological drift.

VI. DISCUSSION: MINIMUM SUFFICIENT MODEL COMPLEXITY

The empirical findings establish an operational principle of **Minimum Sufficient Model Complexity** governed by channel observability and telemetry freshness:

1) *Physics under Fresh Telemetry*: Deterministic aerodynamic curves evaluate pristine telemetry ($\tau = 0$) with asymptotic power conservation ($\partial P / \partial v \propto v^2$ below rated speed, $P \equiv P_{\text{rated}}$ above rated speed). Avoiding out-of-distribution neural variance, they achieve the lowest evaluated surrogate reserve screening cost within the boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0 \text{ m/s}$, $N = 13,883$ events: $589.5\text{k--}881.4\text{k kW} \cdot \text{h}$, compliant $\sim 6.8\%$ violation; Tables~I and~III), where deep representations are superfluous.

2) *Recalibration under Observability*: Under observable laten-

cy (10–30 min), state-conditional recalibration absorbs 55.3% of shortage loss (127.6 $\rightarrow$ 57.0 MWh) and maintains compliant violation rates (7.1%–7.5%) without updating neural weights. Static recalibration breaks down only under contingency horizons ($\tau = 60$ min, Table~A111, 13.8% violation) as cubic extrapolation errors exceed static margins.

3) *Learned Recovery under Pitch Withholding*: Decisive utility emerges when blade-pitch telemetry is withheld (`no_pitch`), where physical rules inflate penalties to 1,378,900 kW $\cdot$ h. The spatio-temporal encoder reconstructs latent regimes from secondary consequence channels (active power divergence, electrical transients, spatial wake advection; 0.302–0.674 N-MI), saving 46.7k–68.6k kW $\cdot$ h.

4) *Posterior Decision Relevance and Baseline Bounds*: The learned posterior does not improve plant-wide reserve screening over a strong wind-speed-conditioned quantile baseline (a +523,044 kW $\cdot$ h penalty, $p = 0.85$); its detectable incremental value is strictly transition-localized. In steady operation, direct wind-speed binning serves as the operational minimum sufficient model, outperforming the posterior by +410,340 kW $\cdot$ h ($p = 0.002$). In dynamic transition windows (± 3 steps), posterior conditioning selectively compresses violation rates (7.24% vs. 8.36%) and reduces shortage exposure (86.6k vs. 90.2k kW $\cdot$ h), confirmed by a significant interaction contrast ($\Delta_{\text{trans}} - \Delta_{\text{steady}} = -296,123$ kW $\cdot$ h, $p < 0.005$). A deployable hybrid policy matches wind-speed binning plant-wide (13.79M vs. 13.78M kW $\cdot$ h, $p > 0.40$). Crucially, dynamic MoE routing provides no measurable advantage over unrouted dense baselines (ratio 1.045, $p = 0.380$; seed delta $< 0.07\%$), while modular regression (STGQ-Modular) matches tail reliability (9.7% violation).

5) *Industrial Deployment Guardrails*: (i) *Spatial value*: spatial-representation value is site-dependent and appears related to exploitable spatial redundancy, not turbine count alone. Wake coupling delivers significant reserve reductions (WTB, Penmanshiel, $p < 0.05$), while sparse layouts (LHB) overfit localized terrain (+43k kW $\cdot$ h penalty); (ii) *Local retraining*: directional transfer asymmetry (Kelmars $\rightarrow$ Penmanshiel NMI 0.77 vs. 0.34) requires local retraining; (iii) *Abstention limits*: selective abstention fails under latency ($p > 0.40$), favoring uniform margin expansion.

VII. LIMITATIONS

We delineate four foundational limitations of this study:

1) *Synthetic Stress*: Impairments are evaluated under synthetic stress (10–30 min backlogs, 60-min latency, Gilbert-Elliott dropouts); field units may exhibit sensor icing or individual pitch actions outside supervisory records.

2) *Risk Surrogacy*: Evaluation operates at Level-1 pre-dispatch operating reserve screening via the PSREI newsvendor surrogate ($\rho = 10$), abstracting Level-2 bulk transmission AC-OPF and balancing settlements.

3) *Horizon Scope*: Benchmarks use a 35-day continuous test split across five random seeds (201–205), omitting multi-annual climatological cycles. Rated wind speeds vary by

model (12.5 m/s on MM92 vs. 14.5 m/s on MM82), requiring OEM calibration.

4) *Deployment Boundaries*: Spatial-representation value remains site-dependent and related to exploitable spatial redundancy rather than turbine count alone; lacking spatial wake coupling (LHB), graph convolutions overfit localized topography (+43k kW $\cdot$ h penalty). Directional transfer asymmetry (Kelmars $\rightarrow$ Penmanshiel NMI 0.77 vs. Penmanshiel $\rightarrow$ Kelmars NMI 0.34) supports local retraining rather than assuming reliable zero-shot transfer. The present study assumes historical blade-pitch availability during offline training, followed by pitch withholding at deployment. Performance in settings where pitch was never historically recorded remains untested.

VIII. CONCLUSION

This study repositions machine learning in reserve screening to a targeted conditional inference mechanism under telemetry degradation. First, under fresh telemetry, deterministic aerodynamic power curves minimize surrogate reserve screening penalties within the boundary band ($|v_{\text{phys}} - 10.5| \leq 1.0$ m/s, $N = 13,883$ events: 589,535 kW $\cdot$ h at $h = 1$, 881,367 kW $\cdot$ h at $h = 6$, $\sim 6.8\%$ violation), collapsing only under unadapted latency (24.0% violation). Second, under observable delays, state-conditional recalibration absorbs 55.3% of shortage loss (127.6 $\rightarrow$ 57.0 MWh). Learned representations acquire utility only when blade pitch is withheld, inferring latent regimes from electromechanical consequences to save 46.7k–68.6k kW $\cdot$ h and double transition recall (0.416 vs. 0.196). Third, dynamic MoE routing confers no statistical advantage over unrouted architectures ($p = 0.380$). Dispatch systems should deploy deterministic rules under clean telemetry, recalibration under observable drift, and learned representations when critical control states are unobservable.

DECLARATIONS

AI Use Statement: The authors utilized OpenAI ChatGPT and Codex to assist with drafting, structural formatting, and prose refinement across manuscript sections and supplementary materials, as well as developing automated audit and consistency-checking scripts. All scientific hypotheses, mathematical formulations, experimental designs, SCADA data processing, model implementations, statistical analyses, causal verification, interpretation of findings, and final approval of the text remain the sole responsibility of the authors.

Code and Data Availability: Raw SCADA datasets (WTB, ENGIE La Haute Borne, Kelmars, Penmanshiel) are public. Code, manifests, derived tables, and checkpoints are provided in the replication repository across declared seeds. The analysis involves no human subjects.

REFERENCES

- [1] J. Dowell and P. Pinson, "Very-short-term probabilistic wind power forecasts by sparse vector autoregression," *IEEE Transactions on Smart Grid*, vol. 7, no. 2, pp. 1063–1070, 2016, doi: 10.1109/TS-G.2015.2424078.
- [2] J. Kruse, E. Cramer, B. Sch"after, and D. Witthaut, "Physics-informed machine learning for power grid frequency modeling," *PRX Energy*, vol. 2, no. 4, p. 043003, 2023, doi: 10.1103/PRXEnergy.2.043003.
- [3] P. Pinson, "Wind energy: Forecasting challenges for its operational management," *Statistical Science*, vol. 28, no. 4, pp. 564–585, 2013, doi: 10.1214/13-sts445.
- [4] R. Doherty and M. O'Malley, "A new approach to quantify reserve demand in systems with significant installed wind capacity," *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 587–595, 2005, doi: 10.1109/TPWRS.2005.846206.
- [5] Y. Wang, F. Zhang, H. Kou, R. Zou, Q. Hu, and J. Wang, "A review of predictive uncertainty modeling techniques and evaluation metrics in probabilistic wind speed and wind power forecasting," *Applied Energy*, vol. 396, p. 126234, 2025, doi: 10.1016/j.apenergy.2025.126234.
- [6] J. G. Slootweg, S. W. H. de Haan, H. Polinder, and W. L. Kling, "General model for representing variable speed wind turbines in power system dynamics simulations," *IEEE Transactions on Power Systems*, vol. 18, no. 1, pp. 144–151, 2003, doi: 10.1109/TPWRS.2002.807113.
- [7] E. Gaertner *et al.*, "Definition of the IEA wind 15-megawatt offshore reference wind turbine," National Renewable Energy Laboratory (NREL), Golden, CO, NREL/TP-5000-75698, 2020. doi: 10.2172/1603478.
- [8] J. Tautz-Weinert and S. J. Watson, "Using SCADA data for wind turbine condition monitoring: A review," *IET Renewable Power Generation*, vol. 11, no. 4, pp. 382–394, 2017, doi: 10.1049/iet-rpg.2016.0248.
- [9] M. A. Ullah, K. Mikhaylov, and H. Alves, "Enabling mMTC in remote areas: LoRaWAN and LEO satellite integration for offshore wind farm monitoring," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 6, pp. 3744–3753, 2022, doi: 10.1109/TII.2021.3112386.
- [10] S. Daenens, T. Verstraeten, P.-J. Daems, A. Nowé, and J. Helsens, "Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data," *Wind Energy Science*, vol. 10, no. 6, pp. 1137–1152, 2025, doi: 10.5194/wes-10-1137-2025.
- [11] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, "Graph WaveNet for deep spatial-temporal graph modeling," in *Proceedings of the 28th international joint conference on artificial intelligence (IJCAI)*, 2019, pp. 1907–1913. doi: 10.24963/ijcai.2019/264.
- [12] S. Guo, Y. Lin, N. Feng, C. Song, and H. Wan, "Attention based spatial-temporal graph convolutional networks for traffic flow forecasting," in *Proceedings of the AAAI conference on artificial intelligence*, 2019, pp. 922–929. doi: 10.1609/aaai.v33i01.3301898.
- [13] L. Bai, L. Yao, C. Li, X. Wang, and C. Wang, "Adaptive graph convolutional recurrent network for traffic forecasting," in *Advances in neural information processing systems*, 2020, pp. 17804–17815.
- [14] J. Park and J. Park, "Physics-induced graph neural network: An application to wind-farm power estimation," *Energy*, vol. 187, p. 115883, 2019, doi: 10.1016/j.energy.2019.115883.
- [15] D. Kim, G. Ryu, and C. Moon, "Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration: A case study of the 400-MW Anholt offshore wind farm," *Applied Energy*, vol. 373, p. 123882, 2024, doi: 10.1016/j.apenergy.2024.123882.
- [16] N. Zehtabiyani-Rezaie, A. Iosifidis, and M. Abkar, "Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability," *PRX Energy*, vol. 2, no. 1, p. 013009, 2023, doi: 10.1103/PRXEnergy.2.013009.
- [17] B. J. Pierre *et al.*, "Design of the Pacific DC Intertie wide area damping controller," *IEEE Transactions on Power Systems*, vol. 34, no. 5, pp. 3594–3604, 2019, doi: 10.1109/TPWRS.2019.2903782.
- [18] G. Ravikumar and M. Govindarasu, "Anomaly detection and mitigation for wide-area damping control using machine learning," *IEEE Transactions on Smart Grid*, vol. 15, no. 6, pp. 5939–5951, 2024, doi: 10.1109/TSG.2020.2995313.
- [19] E. A. Bossanyi, "The design of closed loop controllers for wind turbines," *Wind Energy*, vol. 3, no. 3, pp. 149–163, 2000, doi: 10.1002/we.34.
- [20] F. D. Bianchi, H. De Battista, and R. J. Mantz, *Wind turbine control systems: Principles, modelling and gain scheduling design*. London: Springer, 2006. doi: 10.1007/1-84628-493-7.
- [21] L. Y. Pao and K. E. Johnson, "Control of wind turbines: Approaches, challenges, and recent developments," *IEEE Control Systems Magazine*, vol. 31, no. 2, pp. 44–62, 2011, doi: 10.1109/MCS.2010.939962.
- [22] C. Zhao *et al.*, "A physics-aware dynamic graph and mixture-of-experts framework for wind power forecasting," *International Journal of Electrical Power & Energy Systems*, vol. 176, p. 111769, 2026, doi: 10.1016/j.ijepes.2026.111769.
- [23] J. B. Bremnes, "Probabilistic wind power forecasts using local quantile regression," *Wind Energy*, vol. 7, no. 1, pp. 47–54, 2004, doi: 10.1002/we.107.
- [24] H. A. Nielsen, H. Madsen, and T. S. Nielsen, "Using quantile regression to extend an existing wind power forecasting system with probabilistic forecasts," *Wind Energy*, vol. 9, no. 1–2, pp. 95–108, 2006, doi: 10.1002/we.180.
- [25] N. Shazeer *et al.*, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *International conference on learning representations*, 2017.
- [26] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *Journal of Machine Learning Research*, vol. 23, no. 120, pp. 1–39, 2022.
- [27] J. Zhou *et al.*, "SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array," *Scientific Data*, vol. 11, p. 649, 2024, doi: 10.1038/s41597-024-03427-5.
- [28] H. Hersbach *et al.*, "The ERA5 global reanalysis," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020, doi: 10.1002/qj.3803.